

Lost Person Detection in Aerial Search and Rescue Operations: A Robust Approach Using Optimized YOLO Models

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Abstract—Unmanned Aerial Vehicles (UAVs), or drones, have emerged as a transformative technology for emergency response, particularly in search and rescue operations. Their ability to cover vast and inaccessible areas rapidly makes them ideal for locating missing persons. The integration of artificial intelligence, specifically deep learning-based computer vision, enhances this potential by enabling autonomous, real-time person detection from aerial imagery. This work aims to develop and validate a robust AI-driven system for detecting persons from a drone's perspective. The primary goal is to create a convolutional neural network model that achieves high accuracy and reliability in diverse rural and natural environments, under varying conditions of lighting and flight altitude, while being suitable for deployment on resource-constrained edge devices. The project successfully designed and trained two models based on the YOLOv11 architecture (nano and small versions) on a custom dataset of 4,289 images. The models were trained on a high-performance computing cluster and evaluated on real-world drone footage. The YOLOv11s model demonstrated superior performance, achieving a precision of 0.806 and a recall of 0.760 on one test video, effectively identifying persons in different postures and terrains. While the models showed high efficacy in mAP@50 scores, the stricter mAP@50-95 metric indicated room for improvement in precise localization. The study confirms the viability of using compact YOLO models for real-time person detection in search and rescue scenarios.

Index Terms—Person Detection, Unmanned Aerial Vehicles, YOLOv11, Convolutional Neural Networks, Search and Rescue

I. INTRODUCTION

In recent years, Artificial Intelligence (AI) has experienced exponential growth, transforming numerous sectors through its capacity to analyze vast volumes of data, learn complex patterns, and make autonomous decisions. One domain where artificial intelligence is positioned to exert a substantial influence is in Search-And-Rescue (SAR) operations, wherein the rapid identification of missing individuals may determine the distinction between life and death [1]. The rising frequency and intensity of natural disasters, including earthquakes, wildfires, and floods, have highlighted an urgent requirement for technologies capable of effectively assisting emergency teams in these critical and time-sensitive missions. In this context, news reports regarding the utilization of drones and computer vision algorithms are increasingly prevalent, exemplifying how

automated systems can efficiently cover extensive areas and access terrains that are hazardous or inaccessible to human responders. Consequently, the automatic detection of individuals via aerial imagery has become an essential instrument for the prompt identification of persons in distress.

The integration of Unmanned Aerial Vehicles (UAVs), or drones with advanced AI algorithms, particularly Convolutional Neural Networks (CNNs) and detection models such as You Only Look Once (YOLO), offers a promising solution to enhance the efficacy of rescue operations. This synergy facilitates not only the real-time capture of aerial footage, but also the immediate automated analysis of this visual information, thereby supporting prompt decision-making. Despite notable advancements, significant technical challenges remain. These include ensuring recognition accuracy under adverse environmental conditions, overcoming the computational limitations of embedded devices, and adapting models to unstructured, real-world environments scenarios. These challenges offer a definitive pathway for the research and development of tailored solutions that enhance the application of AI in real-world emergency scenarios.

Our proposal addresses this requirement by developing an intelligent system for person detection from a drone's vantage point. The primary objective is to develop a system capable of recognizing individuals in real-world environments from an aerial perspective using computer vision techniques grounded in artificial intelligence. This instrument is intended for circumstances where direct human intervention proves to be challenging, hazardous, or impractical/inefficient. To attain this overarching objective, several specific aims were established: (1) to design and train a person detection model based on CNNs, capable of accurately identifying humans in aerial imagery under varying lighting, perspective, and background conditions; (2) to validate the system through real-world testing, assessing its performance across different terrains, distances, and environmental factors using metrics such as precision and false-positive rate; (3) to develop a customized, well-labeled dataset that reflects diverse and realistic scenarios; and (4) to provide deployment options for the model on devices with constrained resources, balancing

the trade-off between inference speed and detection accuracy. By leveraging the capabilities of contemporary deep learning and edge computing, this work aims to establish an efficient, functional system applicable in real-world settings, thereby serving as a foundation for future research and humanitarian applications.

The structure of this paper is as follows: Section II reviews related works on the evolution of UAVs technology, the fundamentals of Artificial Neural Networks, as well as Convolutional Neural Networks for object detection. Section III details the system architecture and experimental pipeline, including dataset curation, selection of YOLOv11 variants, and training on a high-performance computing cluster. Section IV presents the model's performance on real-world drone footage, analyzing key detection metrics and qualitative outcomes. Finally, Section V concludes the paper by summarizing the achievements, acknowledging limitations, and proposing directions for future research, such as model deployment and dataset expansion.

II. RELATED WORKS

This work is founded upon three fundamental technological pillars: the progression of UAVs, the theoretical and architectural principles of neural networks, and the advancements in computer vision that facilitate object detection.

The development of UAVs, commonly referred to as drones, has shifted from predominantly military purposes to a broad spectrum of civilian applications [2], [3]. Initially designed for applications such as aerial targets and reconnaissance, contemporary drones have evolved into versatile instruments due to advancements in miniaturization, GPS navigation, and sensor technology. Current civilian applications are diverse, spanning automated package delivery, precision agriculture for crop monitoring and spraying, infrastructure inspection, and, most relevantly, support in emergency and SAR operations [4], [5]. Drones are especially suitable for these missions owing to their capacity to navigate challenging terrains, offer aerial viewpoints, and function in hazardous environments without endangering human lives. Contemporary classifications of UAVs emphasize their diversity, categorizing them by mission (civil/military), propulsion technology (electric, combustion, or turbine), flight altitude, level of autonomy, and airframe configuration (fixed-wing vs. multi-rotor). Multi-rotor platforms are particularly preferred due to their stability and versatility in applications such as aerial cinematography and close-range inspections [6], [7].

The fundamental intelligence that underpins automated detection resides in Artificial Neural Networks (ANNs). ANNs are computational models inspired by the human brain that can learn complex patterns from data [8]. The fundamental processing unit, known as the artificial neuron, converts inputs via a weighted sum followed by a non-linear activation function, such as Sigmoid, Tanh, or the widely adopted Rectified Linear Unit (ReLU), which is crucial for allowing the network to represent non-linear relationships [9], [10]. When organized into multiple layers, these networks are categorized

under Deep Learning (DL). A crucial architecture within deep learning for processing visual data is the CNNs [11]. CNNs are specifically engineered to process data characterized by a grid-like structure, such as images. Their architecture generally comprises convolutional layers that apply learnable filters to extract features such as edges and textures, pooling layers that reduce spatial dimensionality to enhance computational efficiency and achieve translation invariance, and fully connected layers responsible for final classification [12]. This hierarchical framework enables CNNs to acquire progressively intricate features, ranging from basic edges in the initial layers to comprehensive shapes such as human silhouettes in subsequent layers, thereby rendering them highly effective for image recognition tasks.

III. SYSTEM ARCHITECTURE AND EXPERIMENTAL PIPELINE

The development of a robust person detection system for aerial platforms necessitates a comprehensive methodology that addresses both technical challenges and the specific operational considerations of the context. This study concentrates on the development of an artificial intelligence model capable of identifying individuals from an aerial perspective utilizing a drone-mounted camera. Its primary application is intended for search-and-rescue operations in rural and natural environments within the Valencian Community. The critical nature of these applications necessitates a system that is reliable under specific environmental conditions, characterized by a Mediterranean climate with high solar irradiation and semi-arid landscapes, thereby imposing particular robustness and adaptability requirements on the system model. Furthermore, the inherent computational complexity of training deep neural networks with large data volumes requires a high-performance infrastructure to be available to ensure the process feasibility. To address these challenges, a structured methodological pipeline was implemented, comprising the curation and augmentation of a specialized dataset, the selection and optimization of state-of-the-art object detection architectures, and a rigorous training and validation process supported by a specialized GPU-enabled cluster.

A. Dataset Curation and Augmentation

The dataset utilized in our proposal was sourced from the Roboflow platform, which comprises 1,787 images labeled exclusively with the class "human". To mitigate the occurrence of false positives, the dataset incorporates images devoid of individuals, thereby furnishing the model with instances of non-relevant backgrounds. All images have been resized to 640×640 pixels to optimize the balance between detail retention and computational efficiency [13].

The variability within the dataset is notable, encompassing scenarios with diverse lighting conditions, humid and arid landscapes, and semi-urban environments. Furthermore, individuals are observed in various postures, including standing, seated, or recumbent, as illustrated in Figure 1. To enhance the generalization capability of the model, data augmentation

techniques were employed, involving adjustments to parameters such as saturation, brightness, and exposure, thereby simulating various flight and lighting scenarios [14], resulting in 4,289 images overall. The final dataset was partitioned into training (80%) and validation (20%) subsets, adhering to best practices to mitigate overfitting [13].

Dataset preparation adhered to the conventional YOLO framework format [15], optimized for effective detection modeling training. The dataset was organized in a directory hierarchy that segregates images (/images) from their respective annotations (/labels), with further subdivisions designated for training (train), validation (val), and testing (test). Each image, provided in JPEG or PNG format with a resolution of 640×640 pixels, is accompanied by a corresponding text file containing bounding box annotations. These annotations adhere to a standardized format comprising five values per object: the class identifier (in this case, 0, indicating "human"), normalized coordinates of the center (x_center, y_center), as well as normalized dimensions including width and height. All values are articulated within the range [0,1], relative to the image dimensions. The comprehensive dataset configuration was delineated in a data.yaml file, specifying paths to data subsets, the number of classes (nc: 1), class names (names: ['human']), and supplementary Roboflow metadata. This file functions as a crucial interface between the data and the model throughout the training process.

B. Model Selection and Hardware Configuration

Several object detection models were evaluated, including Faster R-CNN, RetinaNet, SSD, and YOLO. Faster R-CNN is a two-stage detector that provides high accuracy; however, it entails substantial computational expenses [16]. RetinaNet incorporates Focal Loss to tackle class imbalance; however, its current ecosystem exhibits less dynamism [17]. SSD is a single-stage model that emphasizes speed but may encounter limitations in small-object detection [18].

YOLOv11 was chosen for its optimal balance between accuracy and efficiency, its contemporary architecture, and the ongoing support from its active development community [15]. YOLOv11 employs a CSP-Darknet-based backbone comprising C3K2, SPPF, and C2PSA, complemented by a neck architecture that integrates multi-scale feature maps to facilitate object detection across varying sizes. These characteristics render it suitable for real-time applications and environments with limited resources.

Model training was performed on the Sirius cluster at the Universitat Politècnica de València, a high-performance system equipped with compute nodes featuring NVIDIA HGX H100 80 GB GPUs, achieving a total capacity of 259 Tflops for general-purpose applications and exceeding 535 Tflops for supercomputing and Artificial Intelligence applications. Specifically, the training process involved requesting 2 NVIDIA HGX H100 GPUs, 48 CPU cores, and 160 GB of RAM from the cluster.

C. Training and Validation Protocol

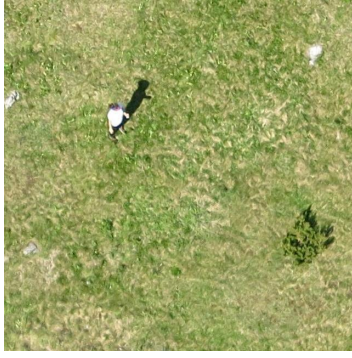
A comprehensive and multi-phase training protocol was implemented to develop, validate, and optimize the detection models, integrating strategies such as transfer learning and early stopping to ensure durable and reliable performance.

1) *Experimental Setup and Hyperparameters* : A virtual environment was established utilizing Conda, wherein essential libraries such as PyTorch, TorchVision, and Ultralytics were installed. Access to GPU nodes was controlled through SLURM scripts specifying the allocated resources. This environment facilitated the execution of training procedures in an isolated and efficient manner. Furthermore, two variants of YOLOv11 were selected: nano (YOLOv11n) and small (YOLOv11s), based on their optimal balance between accuracy and efficiency, as evidenced by the performance of YOLOv11 models trained on the Common Objects in Context (COCO) dataset. The nano model, comprising 2.6 million parameters, is designed for deployment on devices with limited resources, whereas the small model, comprising 9.4 million parameters, provides enhanced performance while maintaining a relatively small size.

2) *Training Regimes and Early Stopping* : The training process was executed using a systematic methodology that integrated transfer learning with advanced optimization techniques. The YOLOv11n and YOLOv11s models were initialized with weights pre-trained on the COCO dataset, thereby utilizing generic visual representations to expedite convergence. The training configuration employed an input image size of 640×640 pixels (imgsz=640) and utilized both GPUs available on the Sirius cluster (device=[0,1]) to optimize computational efficiency. Three distinct training regimens were devised: an initial training phase of 100 fixed epochs, followed by two extended configurations of 800 epochs incorporating early stopping mechanisms with patience values of 20 and 50 epochs, respectively. This design facilitated the assessment of both baseline performance and long-term convergence behavior, with the patience parameter functioning as a safeguard against overtraining by automatically terminating training when no improvement in the validation loss was observed over the specified number of consecutive epochs.

The technical implementation was conducted via SLURM scripts that managed resource allocation within the high-performance cluster. Each experiment produced comprehensive tracking metrics, including loss functions for box regression (box_loss), classification (cls_loss), and focal distribution (dfl_loss), across both training and validation datasets. The selection of these parameters and the training architecture was grounded in well-established practices within the literature [19], thereby facilitating a rigorous comparative evaluation of various configurations and models, while ensuring computational efficiency through optimal resource utilization.

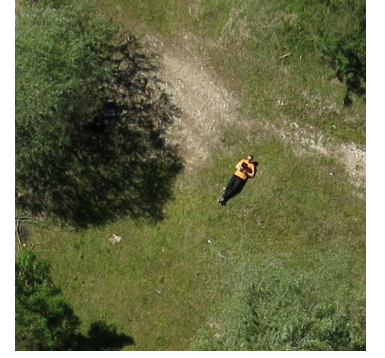
3) *Evaluation Metrics* : The training analysis uncovered a well-defined performance hierarchy and specific convergence patterns, which informed the selection of the most representative results. Figure 2 illustrates the optimal training regime, with the model reaching peak performance at epoch 318 before



(a) Standing



(b) Seated



(c) Recumbent

Fig. 1: Multiple Body Positions Observed.

TABLE I: Training Convergence Analysis for YOLOv11 Models.

Metric	100 Epochs		800 Epochs (p=20)		800 Epochs (p=50)	
	YOLOv11n	YOLOv11s	YOLOv11n	YOLOv11s	YOLOv11n	YOLOv11s
train/box_loss	0.86312	0.72628	0.97943	0.71778	0.48933	0.35799
train/cls_loss	0.46130	0.38862	0.57961	0.42809	0.27184	0.21441
train/dfl_loss	0.95277	0.91707	0.99657	0.90592	0.82767	0.80534
val/box_loss	1.03673	0.92031	0.92022	0.71641	0.70953	0.59342
val/cls_loss	0.50275	0.46411	0.45794	0.34805	0.35761	0.32114
val/dfl_loss	1.02952	1.01183	0.98782	0.91763	0.91674	0.88552

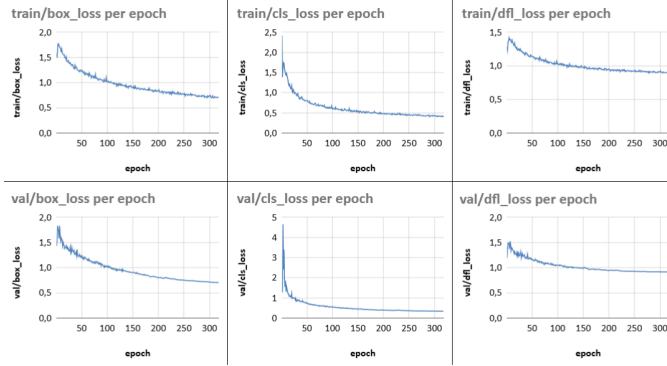


Fig. 2: YOLOv11s Training-Validation Balance (Patience=20).

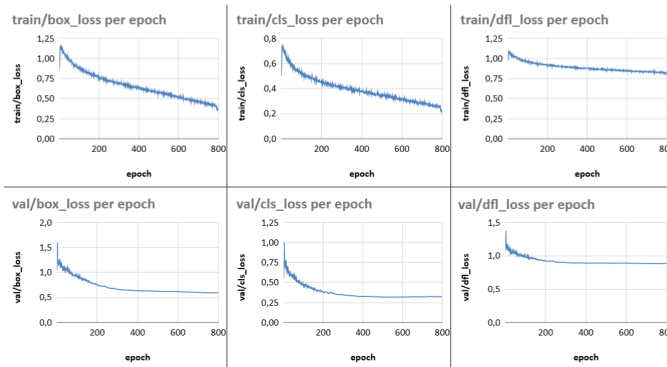


Fig. 3: Performance Saturation in YOLOv11s (Patience=50).

early stopping mechanisms prevent overfitting. In contrast, Figure 3 illustrates the performance ceiling and its associated

risk, as the validation losses plateaued at a near-zero slope, indicating exhausted generalization capacity and the imminent onset of overfitting.

These visual findings are analytically substantiated by Table I, which quantifies the superior performance of the small model across all configurations. Specifically, during the final 800-epoch training, YOLOv11s achieved significantly lower validation losses (box_loss: 0.59342, cls_loss: 0.32114) than YOLOv11n (box_loss: 0.70953, cls_loss: 0.35761), confirming its enhanced learning capability while highlighting diminishing returns and potential overfitting in the extended training phases.

IV. RESULTS

This section presents a comprehensive evaluation of the trained YOLOv11n and YOLOv11s models for person detection in aerial footage. The performance is quantitatively assessed using standard object detection metrics on two real-world videos captured by a drone, providing insight into the models' robustness, accuracy, and practical applicability in varied rural environments.

A. Video Processing and Test Dataset

To evaluate the models under authentic conditions, two separate videos were recorded at the La Llosa aerodrome in Castellón, Spain, using a DJI Air 3S drone, which features a wide-angle 1" CMOS sensor capable of recording 4K/60fps video. The first video (Video 1), lasting about 15 minutes, was captured in a humid environment, characterized by tall and short grasses, while the second video (Video 2), lasting about 20 minutes, was filmed in a drier, more arid region, presenting distinct visual challenges, as illustrated in Figures



Fig. 4: Representative Frames from Video 1.



Fig. 5: Selected Frames from Video 2.

4 and 5. A custom test dataset was meticulously compiled by extracting 600 frames uniformly from both videos, which were then cropped to the model's input dimensions of 640x640 pixels to focus on the central region and avoid additional preprocessing steps.

The extracted frames were then manually annotated using the Roboflow platform to establish ground truth for evaluation purposes. The annotation procedure, as shown in Figure 6, involved delineating the bounding boxes around all individuals within each frame. A primary criterion was that an object was labeled solely if at least 50 percent of it was visible. Frames devoid of individuals were designated as null, thereby producing a meticulously labeled test set that precisely represents the intended operational scenario.

This manually curated dataset, comprising frames from two different environmental contexts, serves as a robust benchmark for assessing the model's generalization capability. This is particularly valuable as it encompasses scenes absent from the initial training dataset, thus offering a rigorous assessment of the model's effectiveness in real-world search and rescue scenarios within the Valencian Community.

The technical specifications of the DJI Air 3S, including its GPS positioning systems and high-resolution video capabilities, ensured the acquisition of high-quality source material. This is critical for computer vision tasks, as the initial image quality directly impacts the potential performance of the detection model. The subsequent manual processing and annotation ensured the dataset's reliability for an equitable and precise performance evaluation.

The concluding phase of dataset preparation involved categorizing the annotations for Video 1 and Video 2, respectively.

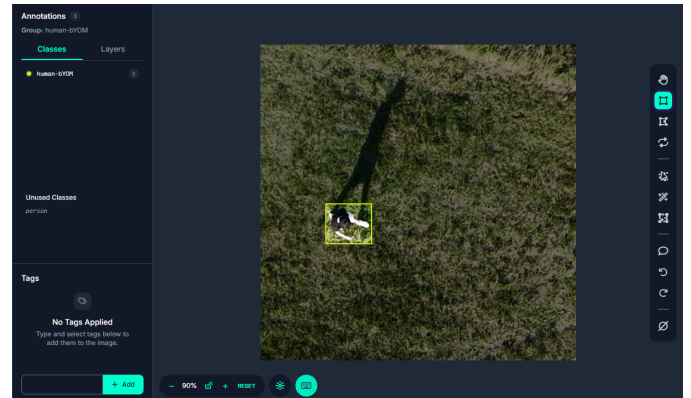


Fig. 6: Manual Labeling Process.

This systematically organized dataset was subsequently used to run inference scripts and derive the performance metrics described in the subsequent subsection.

B. Model Performance and Analysis

The performance metrics of the YOLOv11n and YOLOv11s models, trained over a span of 800 epochs, were systematically assessed using two test videos. The results, which include precision, recall, mAP@50, and the more rigorous mAP@50-95, are shown in Table II.

The examination of Table II discloses two principal findings. Firstly, the YOLOv11s model consistently surpassed the YOLOv11n model in nearly all metrics across both videos, as anticipated due to its greater number of parameters and enhanced computational capabilities. For instance, in Video 1, YOLOv11s achieved a precision of 0.806, compared to 0.736

TABLE II: Comparative performance metrics of YOLOv11 models for aerial person detection.

Metric	Video 1		Video 2	
	YOLOv11n	YOLOv11s	YOLOv11n	YOLOv11s
Precision	0.736	0.806	0.713	0.752
Recall	0.732	0.760	0.724	0.711
mAP@50	0.763	0.784	0.731	0.739
mAP@50-95	0.292	0.307	0.290	0.298

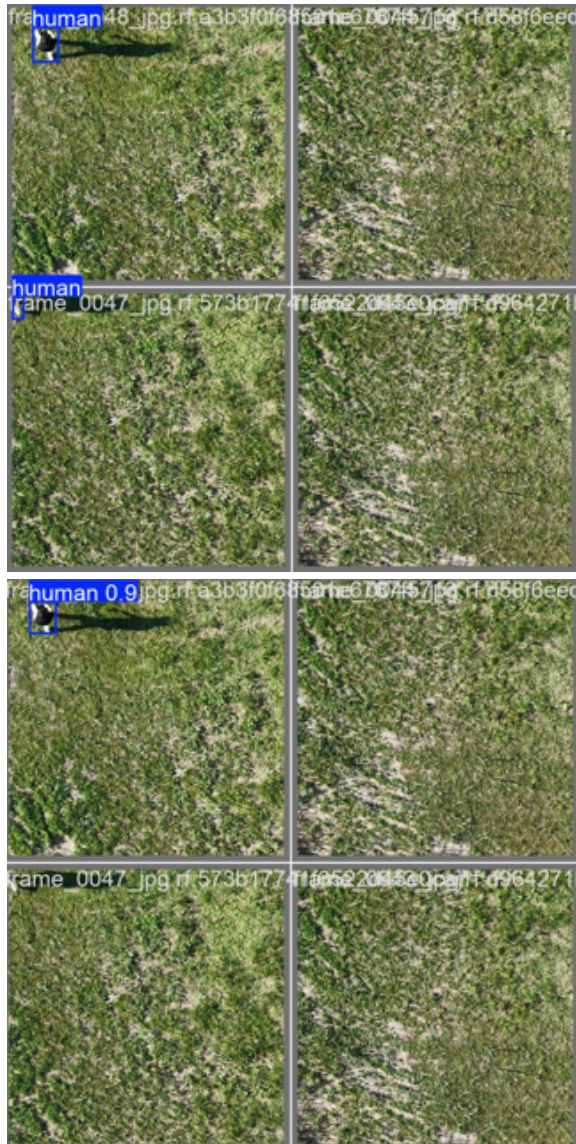


Fig. 7: Qualitative detection results of the YOLOv11n model on Video 1. Ground truth annotations (top) are compared with the model's predictions (bottom).

for the nano model, indicating a lower false-positive rate for the more complex architecture.

Secondly, both models exhibited superior performance on Video 1 compared to Video 2. This discrepancy can be attributed to the composition of Video 2, which primarily consisted of background images devoid of people and incor-

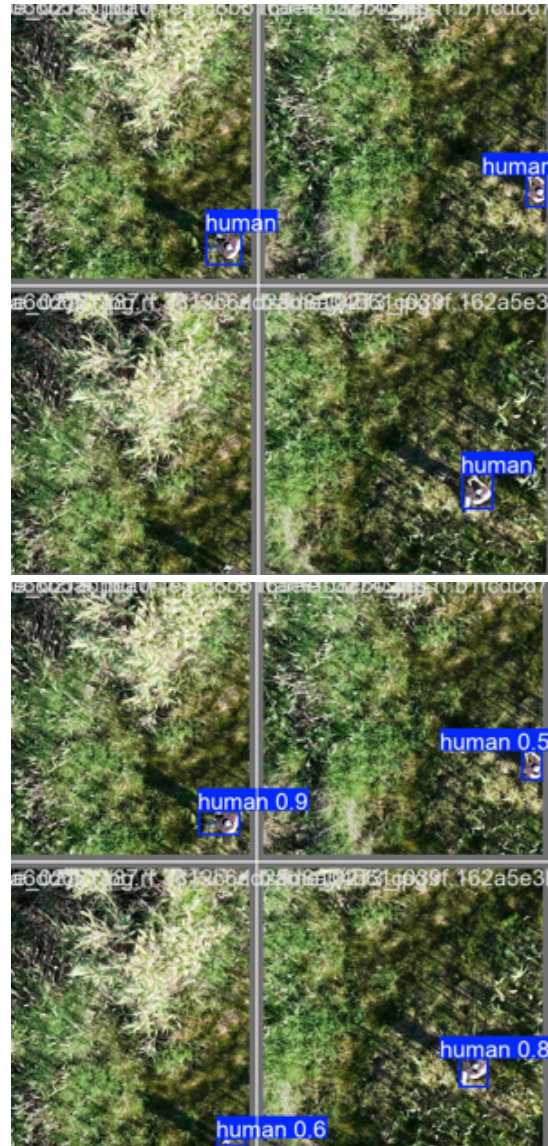


Fig. 8: A Qualitative Comparison of Annotations and Predictions on Video 1. Ground-truth bounding boxes (top) and the detections from the YOLOv11s model (bottom).

porated challenging obstacles, such as tall trees, which could potentially confound the model and consequently adversely affect both precision and recall. Therefore, the more intricate scenery in Video 2 posed a greater challenge for the models to discern true positives from the background.

A notable observation from the results is the significant decline in the mAP@50-95 metric in comparison to mAP@50 for both models. The mAP@50-95 values, representing the average precision over a more rigorous spectrum of Intersection over Union (IoU) thresholds spanning from 0.5 to 0.95, are substantially lower, approximately 0.30. This indicates that, although the models are effective at identifying individuals with a fundamental 50% overlap threshold, their capacity to predict bounding boxes with high spatial precision remains

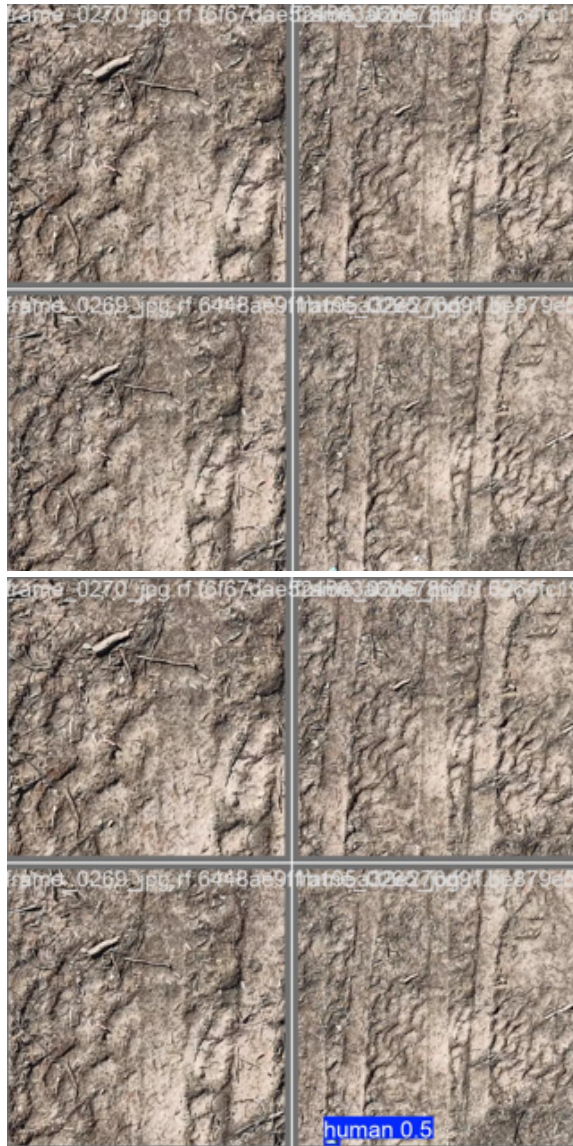


Fig. 9: Qualitative results of the YOLOv11n model on Video 2. Ground truth annotations (top) are compared against the model's predictions (bottom).

constrained, potentially attributable to inconsistencies in the annotation of the training data.

The qualitative analysis of the validation batches, as demonstrated in Figures 7 and 8, corroborates the quantitative findings. A prevalent failure mode observed in both models was their inability to detect persons at the peripheries of the image frame.

Additionally, as depicted in Figure 9, the YOLOv11n model intermittently produced false positives in Video 2, erroneously identifying small, bluish rocks as persons (see wrong prediction on the bottom of the last figure). In contrast, Figure 10 illustrates the superior performance of the YOLOv11s model, which successfully detects a person with a confidence score exceeding 90% in a challenging scenario, and introduces no



Fig. 10: Model Predictions on Video 2. Ground truth annotations (top) and corresponding detections from the YOLOv11s model (bottom).

false predictions despite the many artifacts, thereby emphasizing its increased robustness.

C. Deployment Considerations

For practical implementation, the Ultralytics framework provides comprehensive export options to deploy validated models across various platforms, including ONNX for extensive framework compatibility, TensorRT for optimized inference on NVIDIA hardware, and TFLite for resource-limited edge devices. Critical parameters such as `imgsz`, `half` for FP16 precision, and `int8` for aggressive quantization, can be tuned to balance the trade-off between speed, model size, and accuracy, making the developed models viable for real-time deployment on drone-based systems.

V. CONCLUSIONS

This study successfully achieved its primary objective of developing and training a robust convolutional neural network-based system for person detection from an aerial drone perspective. The work focused on two distinct models, YOLOv11n and YOLOv11s, which demonstrated strong performance across various environmental conditions. The main findings indicated that the YOLOv11s model consistently outperformed the nano variant, attaining a precision of 0.806 and an mAP@50 of 0.784 on validation data, thereby illustrating a superior balance between accuracy and computational efficiency. The models were subjected to rigorous validation utilizing a manually assembled, real-world drone video dataset that was entirely excluded from the training dataset, thus confirming their capacity for generalization. However, challenges were identified in detecting objects at the peripheries of images and a tendency for false positives in cluttered backgrounds. Additionally, a significant contribution of this research is the development and public dissemination of a comprehensive and annotated image dataset, augmented to enhance diversity, which will serve as a valuable resource for future studies. Finally, the project delineates specific strategies for implementing these optimized models on resource-constrained edge devices, thereby satisfying the objective of practical deployment.

Building upon the findings of this study, several promising avenues for future research emerge. The immediate next step involves the practical deployment of the trained models onto an operational drone system to assess their performance and latency within a fully functional, hardware-constrained environment, thereby surpassing offline video analysis. To further enhance efficiency in such deployments, it is recommended that techniques such as model quantization and pruning be explored to decrease computational costs and memory requirements of the most effective models, without significant compromise to accuracy. Additionally, expanding the dataset to include a wider range of scenarios – such as diverse terrains, severe occlusions, and adverse weather conditions – would be instrumental in addressing the identified limitations, and in improving the robustness and reliability of the models for real-world search-and-rescue operations.

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